DSAA3071 Week 5 Candidate-v3.2 Development Note

Development split only. held-out not run; not a final accuracy claim.

Technical summary. C32 is the strongest development condition so far on aggregate MAE: question MAE improves from `R1=2.614` to `C32=2.557`, and total MAE improves from `R1=20.143` to `C32=14.429`. The result is promising but not final: Q8 remains the main unresolved error source, Q6 worsens versus R1, and C32 still has slightly higher severe-error rate than R1.

What Was Tested

1. Same data

Seven anonymous development students from the reviewed DSAA3071 Week 5 transcript snapshot.

2. Same gold

Official per-question scores from `primary_scores.csv`; no official rationales are available.

3. New package

C32 uses `candidate-v3.2` plus `rubric_v2.json` to reduce over-harsh grading on Q7-Q9.

This note covers `DSAA3071` / `week5_test` on branch `codex/dsaa3071-week5-dev-transcripts` at analysis commit `7f4e06a`.

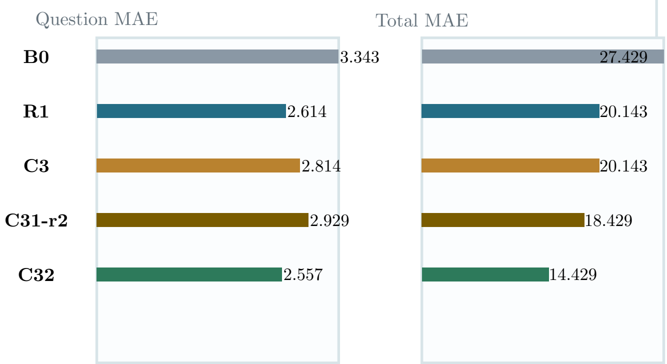
Reproducibility Anchors

Anchor	Value
Metric record	`experiments/records/DSAA3071-week5-candidate-v31-dev-plan/dev-metrics-deepseek-c32.json`
Run protocol	`experiments/records/DSAA3071-week5-candidate-v31-dev-plan/RUN-PROTOCOL-C32.md`
Prompt snapshot	`experiments/records/DSAA3071-week5-candidate-v31-dev-plan/prompts/grade_candidate_v3_2_strict_schema.txt`
Rubric	`experiments/records/DSAA3071-week5-prep/rubric_v2.json`
Skill snapshot	`experiments/skill_versions/skill_candidate_v3_2.json`
Data snapshot	`95e744f5811d...`
Text source hash	`9f9be72bf088...`

Key Findings With Visual Evidence

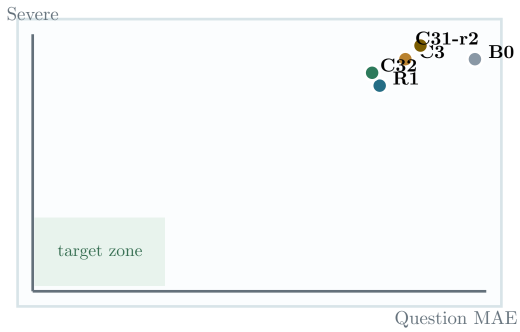
Best question MAE	Best total MAE	Exact agreement	Severe error
2.557	**14.429**	**52.9%**	**22.9%**
C32, lower is better	C32, lower is better	C32 ties C31-r2	C32, still above R1

Aggregate MAE comparison



Shorter bars are better. C32 has the lowest question MAE and total MAE on the development split.

Accuracy-risk map



Lower left is better. C32 moves left versus R1 but remains slightly above R1 on severe-error rate.

Condition Details

Cond.	Description	Validation	Tokens	Q MAE	Total MAE	Severe
B0	baseline prompt + rubric v0	passed (7/7)	63448	3.343	27.429	24.3%
R1	baseline prompt + rubric v1	passed (7/7)	92633	2.614	20.143	21.4%
C3	candidate-v3 prompt + rubric v1	passed (7/7)	104803	2.814	20.143	24.3%
C31-r2	candidate-v3.1 r2 open-ended adequacy prompt + rubric v1	passed_after_recovery (7/7)	107658	2.929	18.429	25.7%
C32	candidate-v3.2 prompt + rubric v2	passed (7/7)	116758	2.557	14.429	22.9%

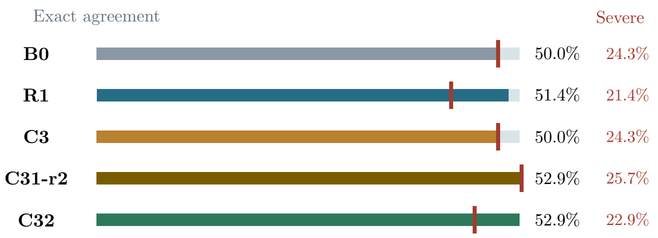
C32 improves aggregate MAE while preserving the same seven-student development split. This table should be read as calibration evidence, not as a final leaderboard.

Per-Question Error Concentration

Question	B0	R1	C3	C31-r2	C32	Best
Q5	7.000	4.857	5.857	2.571	2.714	C31-r2
Q6	2.143	1.714	1.857	2.714	4.429	R1
Q7	4.429	2.429	4.429	5.000	2.429	R1, C32
Q8	3.857	5.714	4.857	5.000	5.571	B0
Q9	14.857	9.714	9.714	11.714	9.000	C32
Q10	1.143	1.714	1.429	2.286	1.429	B0

The strongest C32 improvements are Q7 and Q9 relative to C31-r2. The biggest blocker is still Q8: C32 Q8 MAE is `5.571`, almost unchanged from R1 and worse than B0.

Exact agreement and severe-error bars



Colored bars show exact agreement. Red ticks and red labels show severe-error rate.

C32 Q7-Q9 Development Error Table

Student	Question	Gold	C32	Error
S017	Q7	20.000	17.000	−3.000
S017	Q8	9.000	5.000	−4.000
S017	Q9	25.000	12.000	−13.000
S021	Q7	15.000	16.000	1.000
S021	Q8	0.000	5.000	5.000
S021	Q9	12.000	10.000	−2.000
S002	Q7	10.000	15.000	5.000
S002	Q8	10.000	3.000	−7.000
S002	Q9	25.000	25.000	0.000
S015	Q7	10.000	13.000	3.000
S015	Q8	10.000	4.000	−6.000
S015	Q9	25.000	17.000	−8.000
S020	Q7	18.000	18.000	0.000
S020	Q8	10.000	1.000	−9.000
S020	Q9	10.000	2.000	−8.000
S016	Q7	0.000	0.000	0.000
S016	Q8	8.000	5.000	−3.000
S016	Q9	25.000	5.000	−20.000
S022	Q7	10.000	5.000	−5.000
S022	Q8	10.000	5.000	−5.000
S022	Q9	25.000	13.000	−12.000

This table uses anonymous IDs only and does not copy student answer text. It shows why the result is still mixed: Q7 is mostly controlled, Q9 improved but remains under-scored for several students, and Q8 remains unstable.

Metric Definitions And Method

- `question_score_mae`: mean absolute error over all student-question pairs. Lower is better.
- `normalized_question_score_mae`: mean absolute error divided by question max score. Lower is better.
- `question_exact_agreement`: fraction of student-question pairs where model score exactly equals official score. Higher is better.
- `severe_error_rate_abs_ge_5`: fraction of student-question pairs with absolute error at least 5 points. Lower is better.
- `total_score_mae`: mean absolute error between model total and official total per student. Lower is better.

The model run used DeepSeek public API, `deepseek-v4-pro`, text-only inputs, the reviewed transcript snapshot, and packet `C32-dev-reviewed-r1`. The model output schema validation passed for all seven development students.

Limitations And Next Step

- held-out not run; this is not a final accuracy claim.
- The development split has only seven anonymous students.
- Official scores provide per-question numbers but not detailed official rationales.
- C32 changes rubric, prompt, and skill together by design, so it evaluates the calibrated package rather than one isolated factor.
- Next step: diagnose Q8 specifically before freezing the skill or running a held-out test.